TRIVAPT - Architecture Overview

India's First AI-Powered Web Application VAPT Toolkit

Developed by Trident RedOps

1. Architectural Mission

TRIVAPT is architected to **replicate, enhance, and automate the complete decision-making workflow of experienced VAPT professionals** through an intelligence-driven, phase-oriented system design.

Traditional assessments rely on analysts manually selecting tools, interpreting outputs, filtering noise, correlating findings, and deciding what to test next. TRIVAPT transforms this human-intensive process into a **structured, automated methodology pipeline** that mirrors how real security assessments are performed in practice.

Instead of executing tools independently, TRIVAPT creates a **continuously expanding understanding of the target environment**. Each phase of execution feeds meaningful intelligence into the next phase, enabling deeper exploration of the attack surface with every step.

The architecture is designed so that:

- Discovery leads to contextual analysis
- Analysis leads to intelligent testing
- Testing produces correlated evidence
- Evidence becomes structured knowledge for AI interpretation

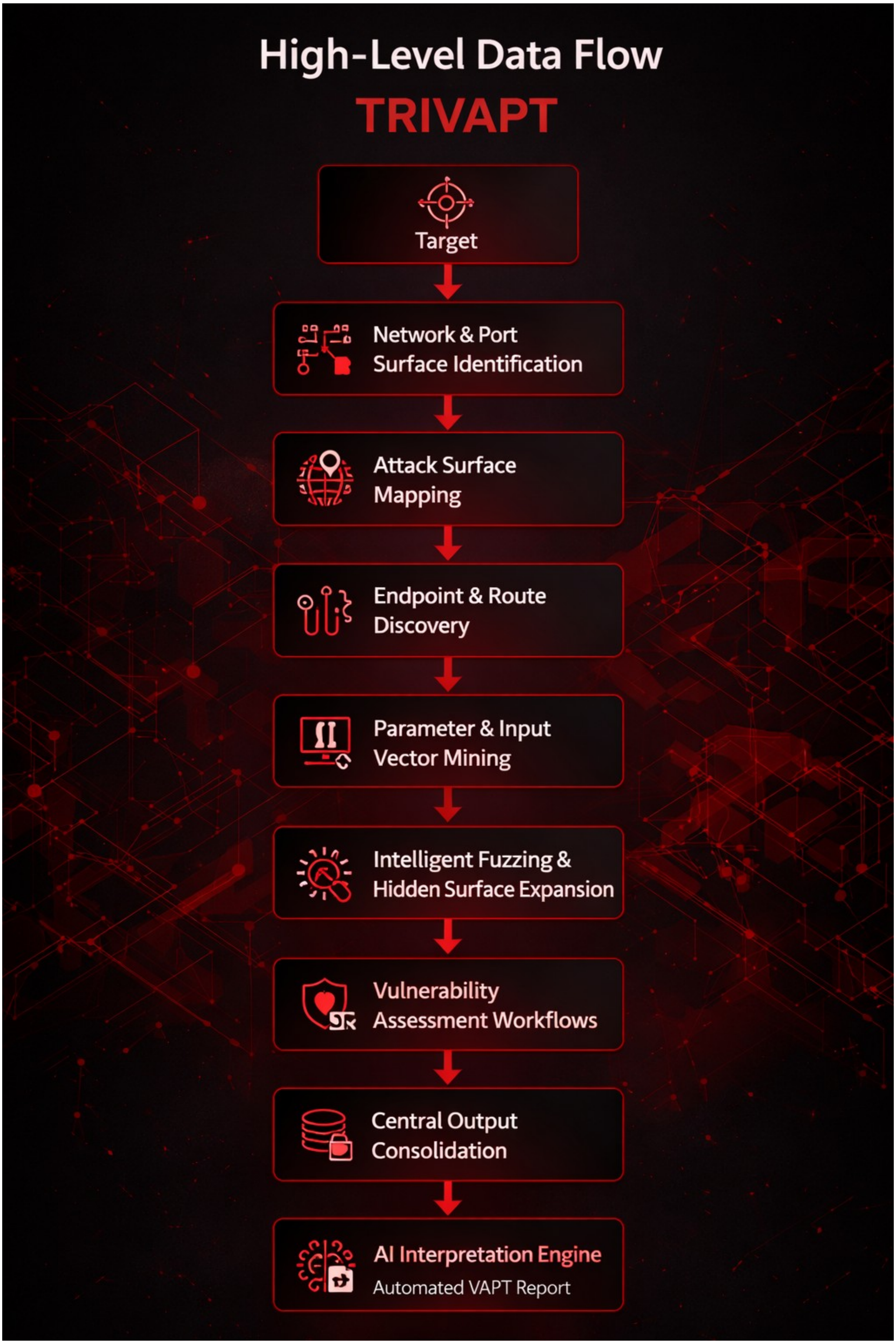
This dataset is then processed by an AI model that interprets the relationships between assets, endpoints, parameters, behaviors, and findings to produce a **professional, structured Vulnerability Assessment report** automatically.

TRIVAPT's mission is not merely to automate scanning -
it is to **automate VAPT thinking**.

2. Foundational Design Principles

- **Methodology-first execution**
- **Phase-wise attack surface expansion**
- **Stateful data reuse across phases**
- **Centralized output consolidation**
- **AI interpretation of complete assessment data**

3. High-Level Data Flow



4. Phase 0 - Network & Port Surface Identification

TRIVAPT begins by identifying how the target is exposed at the network level.

- Live host detection
- Open port discovery
- Service fingerprinting
- Exposure mapping of web-reachable services

This ensures web assessment is aligned with actual service exposure.

5. Phase 1 - Attack Surface Mapping

- Discovery of domains, subdomains, and related assets
- Identification of externally reachable application instances
- Establishment of complete assessment scope

6. Phase 2 - Endpoint & Application Route Discovery

- Collection of live URLs and application paths
- Identification of hidden, unlinked, and archived routes
- Structural mapping of the application

7. Phase 3 - Parameter & Input Vector Mining

- Identification of user-controlled inputs across endpoints
- Detection of injectable parameters
- Expansion of potential attack entry points

8. Phase 4 - Intelligent Fuzzing & Surface Expansion

- Context-aware fuzzing across discovered routes
- Discovery of undocumented functionalities
- Continuous enlargement of the attack map

9. Phase 5 - Vulnerability Assessment Workflows

TRIVAPT performs automated vulnerability testing driven by discovered context:

- Input validation weaknesses
- Access control issues
- Misconfigurations
- Exposure of sensitive functionalities
- Logical endpoint behavior analysis

10. Central Output Consolidation Engine

A core architectural component of TRIVAPT:

- All outputs from every phase are automatically saved
- Data is normalized and merged into a **single consolidated text dataset**
- This dataset represents the complete technical understanding of the target

No output is discarded. Everything becomes intelligence.

11. Intelligence Layer

TRIVAPT includes a decision layer that:

- Determines what to scan next based on previous findings
- Eliminates redundant operations
- Prioritizes meaningful targets
- Mimics expert VAPT analyst reasoning

12. AI Interpretation & Report Generation

The consolidated dataset is processed by a dedicated AI model that:

- Understands relationships between assets, endpoints, parameters, and behaviors
- Identifies vulnerability patterns
- Generates a structured, professional Vulnerability Assessment report including:
    - Affected assets and endpoints
    - Risk explanation
    - Technical evidence
    - Remediation guidance

13. Architectural Strengths

- Eliminates manual recon and output filtering
- Prevents human error in phase chaining
- Continuously expands attack understanding
- Produces consistent, analyst-grade reports
- Functions as an automated VAPT analyst

14. Extended Architectural Capabilities

TRIVAPT architecture inherently supports:

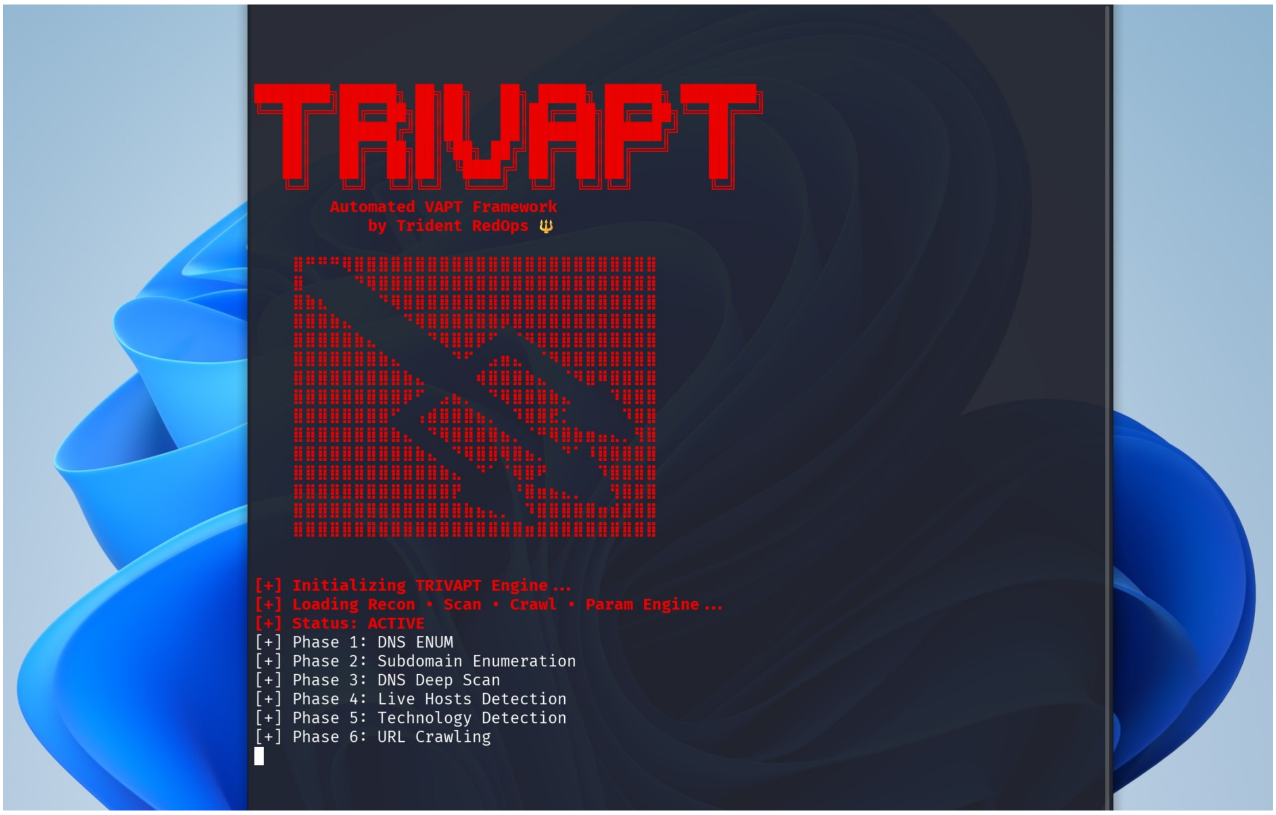
- Deep web application assessment
- Complex multi-endpoint analysis
- Context-aware vulnerability detection
- Large-scale assessment across web services

15. Future Scope Embedded in the Architecture

The architecture is designed to seamlessly extend into:

- API security assessment
- Authentication and session analysis
- Business logic vulnerability detection
- Cloud application surface mapping
- Continuous VAPT monitoring mode
- Learning-based AI improvements from previous assessments

These extensions integrate into the same pipeline without architectural changes.



16. Conclusion

TRIVAPT is India's first AI-powered Web Application VAPT toolkit purpose-built to transform how Vulnerability Assessment and Penetration Testing is executed, interpreted, and reported.

Unlike conventional approaches that rely on running isolated tools and manually interpreting scattered outputs, TRIVAPT delivers an **intelligence-driven, methodology-centric assessment pipeline**. It continuously expands its understanding of the target by chaining network discovery, attack surface mapping, endpoint analysis, input vector mining, intelligent fuzzing, and vulnerability assessment into a single cohesive workflow.

Every artifact generated during assessment is **preserved, correlated, and consolidated** into a unified intelligence dataset. This dataset is then processed by TRIVAPT's AI interpretation engine, which converts raw technical evidence into **context-aware Vulnerability Assessment insights** and a **professionally structured VAPT report**.

TRIVAPT does not merely automate scanning — it automates the **thinking process of a VAPT analyst**.

By combining:

- Network & port surface identification
- Deep attack surface exploration
- Hidden endpoint and parameter discovery
- Intelligent fuzzing and vulnerability workflows
- Centralized output intelligence
- AI-based VA report generation

TRIVAPT redefines VAPT from a collection of tools into a **self-orchestrating security assessment system**.

This positions TRIVAPT as a pioneering innovation from India in the field of AI-driven cybersecurity automation — executing **methodology, intelligence, and reporting** as a single unified engine.

Built in India. Built for the Future of Cybersecurity.